

1) Movie Quote Normalizer

Goal: Write `normalize_quote(text) -> list[str]` that lowercases, strips punctuation, collapses spaces, and splits into words.

Imports: `re`

Edge cases: If text is empty/whitespace, raise `ValueError`.

Example:

" May THE Force—be with you!! " →
["may", "the", "force", "be", "with", "you"]

Hint: `re.sub(r"[^\w\s]", "", ...)` and `split()`.

2) Batch Filename Slugger

Goal: Given a list of filenames, produce safe “slugs” for saving to disk.

Imports: `pathlib`

Rules:

- Lowercase, replace spaces with `-`, drop non-alphanumerics (keep `-` and `.`).
- If a slug collides, append `-2`, `-3`, ...
- If input isn't a string, **skip** and record an error count.

Example:

["My Photo.JPG", "my-photo.jpg", "Notes (Final).txt", 42] → ["my-photo.jpg", "my-photo-2.jpg", "notes-final.txt"] and `errors=1`.

Hint: Use a set to track seen slugs; `Path(name).suffix`.

3) CSV-ish Number Parser

Goal: Parse a multiline string of comma-separated integers to a running **sum**.

Imports: `none` (or `io.StringIO` if you like)

Rules: Ignore blank lines. If a line contains any non-int token, **log it** and continue (don't crash).

Example:

swift

CopyEdit

```
"10,20\nx,2\n5\n"
```

Output: sum=37, bad_lines=1.

Hint: try: int(token) except ValueError: inside a loop.

4) Simple Password Checker

Goal: check_password(pw) returns True if: length ≥ 8 , has upper, lower, digit, and one of !@#\$%^&*.

Imports: re (optional)

Exceptions: If pw isn't str, raise TypeError.

Example: "Navy7!cat" $\rightarrow$ True.

Hint: Either multiple any(c.is...) checks or regexes.

5) Random Sentence Mixer

Goal: Build make_sentence(n=5) that picks random words from three lists (subjects, verbs, objects) to form n sentences.

Imports: random

Rules: Capitalize first word, end with .

Example: ["Robots compile dreams.", "Algorithms sip coffee."]

Hint: random.choice, str.capitalize().

6) Subtitle Word Frequencies

Goal: Given raw .srt text, strip timestamps/index lines and count top **N** words.

Imports: re, collections.Counter

Rules: ignore numbers like 00:01:02,345, drop punctuation, lowercase.

Example: returns [("yeah", 12), ("okay", 9), ...]

Hint: Remove lines containing -->, then normalize as in #1.

7) Safe JSON Activity Log

Goal: Append an event dict to a JSON file that stores a **list** of events. Create the file if missing.

Imports: json, datetime

Exceptions:

- If file exists but is corrupt JSON, back it up to .bak and start fresh.
- If event is missing required keys, raise ValueError.

Example event: {"type": "click", "user": "alice"} augmented with ISO timestamp.

Hint: Catch json.JSONDecodeError; write atomically with a temp file if you want a stretch.

8) Caesar Cipher (with Validation)

Goal: Implement encrypt(text, shift) and decrypt(text, shift) for letters A–Z/a–z; keep punctuation unchanged.

Imports: none

Exceptions:

- Non-int shift → TypeError
- Non-str text → TypeError

Example: encrypt("Hello, World!", 3) → "Khoor, Zruog!"

Hint: Use ord/chr and modulus within ranges.

9) URL Domain Tally

Goal: Given a list of URLs, return a dict of {domain: count}.

Imports: urllib.parse

Exceptions: Skip and count invalid URLs (no scheme/host).

Example:

```
["https://example.com/a","http://example.com","https://sub.example.com"  
"] → {"example.com": 2, "sub.example.com": 1}  
Hint: urlparse(u).hostname; watch for None.
```

10) Mini Retry Wrapper

Goal: Implement `retry(times=3, exceptions=(Exception,))` decorator to retry a function on specific exceptions with small delays.

Imports: `time`, `functools`

Behavior: Exponential backoff: 0.1s, 0.2s, 0.4s...; re-raise after last attempt.

Example: Wrap a flaky function that raises `ValueError` half the time.

Hint: Use `functools.wraps` and a for loop; catch only the allowed exceptions.